

# Energy Homework 1

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Directions:** For each scenario:

1. Define the system.
  2. Choose a reference point where  $h = 0$ .
  3. Identify energy types at the listed instants.
  4. Draw the energy pie charts.
- 

1. **Ball tossed upward and caught at the same height it was thrown at (ignore air resistance).**

What is the system? Is it open or closed?

open closed

What is the reference point? Is there one?

Instants:

- At release
- At the highest point
- Getting caught again

Energy types present at each instant (circle all that apply):

- At release:

K    $U_g$     $U_s$

- At the highest point:

K    $U_g$     $U_s$

- Getting caught again:

K    $U_g$     $U_s$

Draw energy pie charts:

2. **An object is dropped from a balcony (ignore air resistance).**

What is the system? Is it open or closed?

open closed

What is the reference point? Is there one?

Instants:

- At release
- Halfway down

- Hitting the ground

Energy types present at each instant (circle all that apply):

- At release:
- Halfway down:
- Hitting the ground:

|   |       |       |
|---|-------|-------|
| K | $U_g$ | $U_s$ |
| K | $U_g$ | $U_s$ |
| K | $U_g$ | $U_s$ |

Draw energy pie charts:

### 3. Cart launched by a spring on a level track (frictionless).

What is the system? Is it open or closed?

open closed

What is the reference point? Is there one?

Instants:

- Spring fully compressed
- Spring partially compressed
- Spring relaxed, cart moving

Energy types present at each instant (circle all that apply):

- Fully compressed:
- Partially compressed:
- Moving freely:

|   |       |       |
|---|-------|-------|
| K | $U_g$ | $U_s$ |
| K | $U_g$ | $U_s$ |
| K | $U_g$ | $U_s$ |

Draw energy pie charts:

### 4. Skater gliding on level ground (constant speed, frictionless).

What is the system? Is it open or closed?

open closed

What is the reference point? Is there one?

Instants:

- Early in motion
- Later in motion

Energy types present at each instant (circle all that apply):

- Early:

K    $U_g$     $U_s$

- Later:

K    $U_g$     $U_s$

Draw energy pie charts:

Why does  $U_g$  not change in this situation?